A standard US Letter writing document is **612 points (pt) wide** (8.5 inches). [1]
Because 1 typographic point equals exactly 1/72 of an inch, standard document widths convert directly as follows: [1]

Document Format	Standard Width (in / mm)	Width in Points (pt)	Printable Text Width (with 1-inch margins)
US Letter (North America standard)	8.5 in (215.9 mm)	**612 pt**	**468 pt** (6.5 in)
A4 (International standard)	8.27 in (210 mm)	**595 pt** (595.28 pt exact)	**451 pt** (6.27 in)
US Legal	8.5 in (215.9 mm)	**612 pt**	**468 pt** (6.5 in)

If you are setting up a layout in design software, CSS print stylesheets, or a PDF generator, check the full **PrintReadyKit point conversion table** for exact fractional values across other paper formats. [1]
Are you measuring in points for a specific project? Let me know if you are:

- Configuring **custom page margins or columns** in software like Word, Google Docs, or InDesign
- Setting up a **PDF export or CSS print stylesheet**
- Calculating **line lengths** for a specific font size

I can help you determine the exact width or column measurements you need.

For the PNG download, the width of the image should be 612 pt.
468 pts of the area is where the text fits in.
The minimum height is 792 pt.
648 pt of the height is printable. There are one-inch margins.
The border is outside of the text area.

With the current font, how many lines will fit on one page?
What are the dimensions in pixels?
Maybe we can fit 20 pages high per message, but there is also a 15,000 character limit.

Here is the exact breakdown of dimensions, line capacity, and page height calculations based on standard US Letter geometry (8.5″ × 11″ at 72 points per inch).

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1. Dimensions in Pixels

Points (`pt`) are physical typography units ($1\text{ in} = 72\text{ pt}$). In digital PNG images, the pixel dimensions depend on the target resolution (DPI):

Resolution Standard	**Scale Ratio**	**Image Dimensions (Width × Min Height)**	**Printable Text Area (468 pt × 648 pt)**	**1-Inch Margins (72 pt)**
72 DPI (1 pt = 1 px nominal)	1.0 px / pt	**612 × 792 px**	**468 × 648 px**	72 px
96 DPI (Standard Web / CSS)	1.333 px / pt ($4/3$)	**816 × 1,056 px**	**624 × 864 px**	96 px
144 DPI (2× HiDPI / Retina)	2.0 px / pt	**1,224 × 1,584 px**	**936 × 1,296 px**	144 px
300 DPI (Print Publication Quality)	4.167 px / pt	**2,550 × 3,300 px**	**1,950 × 2,700 px**	300 px

Recommendation:

- **96 DPI (816 × 1,056 px)** is standard for 1:1 web viewing and desktop screens.
- **144 DPI (1,224 × 1,584 px)** is ideal for crisp, razor-sharp text on modern displays (and fits cleanly within 32,000 px for up to 20 pages).

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2. How Many Lines Fit on One Page?

The printable vertical space inside the 1-inch margins is **648 pt**.

Depending on which font sizing is used:

1. **Using Print Stylesheet Font (`10 pt`, `1.4` line-height = 14 pt line pitch)**: $$ \frac{648\text{ pt}}{14\text{ pt/line}} \approx \mathbf{46\text{ lines per page}}$$ *(If reserving space on Page 1 for the card header (~42 pt) and footer (~28 pt), Page 1 fits **~41 body lines**, and continuation pages fit **46 lines**).*
2. **Using the Current Canvas Font (`16px`, `line-height = 24`)**:
    - **At 72 DPI** (24 pt line pitch): $$\frac{648\text{ pt}}{24\text{ pt/line}} = \mathbf{27\text{ lines per page}} \quad (\approx 23\text{ lines on Page 1 with header/footer})$$
    - **At 96 DPI CSS Pixels** (864 px text height, 24 px line pitch): $$\frac{864\text{ px}}{24\text{ px/line}} = \mathbf{36\text{ lines per page}} \quad (\approx 31\text{ lines on Page 1 with header/footer})$$

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3. Will a 15,000-Character Message Exceed 20 Pages?

No, it will easily fit within 20 pages.

- **Characters per line**: In a 468 pt line width, standard monospace typography fits **~65 to 78 characters per line** (equivalent to a classic 80-column page).
- **Total lines for 15,000 characters**:
    - Full-density prose (~70 chars/line): $15,000 / 70 \approx \mathbf{214\text{ lines}}$.
    - Shorter lines / code / paragraphs (~45 chars/line): $15,000 / 45 \approx \mathbf{333\text{ lines}}$.
- **Total pages needed**:
    - At **46 lines/page** (10 pt font): 214 lines $\rightarrow$ **~5 pages** (or ~7 pages with short lines).
    - At **36 lines/page** (16px @ 96 DPI): 214 lines $\rightarrow$ **~6 pages** (or ~9 pages with short lines).
    - At **27 lines/page** (24 pt pitch): 214 lines $\rightarrow$ **~8 pages** (or ~12 pages with short lines).

Even at maximum character count with frequent line breaks, a 15,000-character message will only reach **6 to 12 pages**, well under the 20-page threshold.

Next Step

Would you like me to update `downloadMessageAsPng` in index.php now with these exact dimensions (612 pt width, 468 pt text column, 1-inch margins, minimum 792 pt height, with the border enclosing the text area)? If so, please let me know your preferred pixel scale: **96 DPI (816 × 1056 px)** or **144 DPI (1224 × 1584 px)**.

First I'd like to know what the information is for 144 DPI with the font at 12pt.

Here is the complete specification for **144 DPI with a 12pt font**:

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1. Pixel Dimensions & Geometry (144 DPI = 2.0 px per pt)

At 144 DPI, exactly **$2\text{ pixels} = 1\text{ point}$**:

Section	Size in Points (`pt`)	Size in Pixels (`px`)	Physical Size (Inches)
Total Image Width	612 pt	**1,224 px**	8.5 in
Minimum Height (1 Page)	792 pt	**1,584 px**	11.0 in
1-Inch Margins (all 4 sides)	72 pt each	**144 px each**	1.0 in each
Printable Text Area Width	468 pt	**936 px**	6.5 in
Printable Text Area Height	648 pt	**1,296 px**	9.0 in
Enclosing Border (outer boundary)	$468\text{ pt} \times 648\text{ pt}$	**$936\text{ px} \times 1,296\text{ px}$**	Placed at $X=144\text{ px}, Y=144\text{ px}$

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2. Font Sizing & Line Capacity (12pt Font)

In 144 DPI graphics, a **12pt font renders at 24 px** (12×2):

Line Pitch / Leading	Font & Line Height	Lines per Full Page (648 pt / 1,296 px)	Lines on Page 1 (with Header & Footer)
Comfortable (1.5× leading)	12pt font, **18pt pitch** (36 px)	**36 lines**	**~31 lines**
Standard (1.33× leading)	12pt font, **16pt pitch** (32 px)	**40 lines**	**~35 lines**
Compact (1.25× leading)	12pt font, **15pt pitch** (30 px)	**43 lines**	**~37 lines**

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3. Characters per Line (in 468 pt / 936 px Width)

- **Monospace Font** (SF Mono, Menlo, Courier):
  - Each character occupies $7.2\text{ pt} = \mathbf{14.4\text{ px}}$.
  - Character capacity: $\frac{936\text{ px}}{14.4\text{ px}} = \mathbf{65\text{ characters per line}}$. *(65 characters per line aligns with standard book and typography readability guidelines of 60–75 characters).*
- **Proportional / Sans Font** (System UI, Helvetica, Arial):
  - Average character width is $\approx 12.5\text{ px}$.
  - Character capacity: $\approx \mathbf{75\text{ characters per line}}$.

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4. How 15,000 Characters Fit Across 20 Pages

1. **Lines needed for 15,000 characters**:
   - Normal paragraph prose (~65 characters/line): $15,000 / 65 \approx \mathbf{230\text{ lines}}$.
   - Short lines, spacing, or code (~40 characters/line): $15,000 / 40 \approx \mathbf{375\text{ lines}}$.
2. **Total Pages Needed**:
   - At **36 lines/page** (18pt pitch):
     - Prose: **~6.5 pages** (7 pages total)
     - Short lines: **~10.5 pages** (11 pages total)
   - At **40 lines/page** (16pt pitch):

    - Prose: **~5.9 pages** (6 pages total)
    - Short lines: **~9.5 pages** (10 pages total)
3. **Maximum 20 Pages Height Limit**:
    - 1 Page: $1,584\text{ px}$ (792 pt)
    - **20 Pages Maximum**: $20 \times 1,584\text{ px} = \mathbf{31,680\text{ px}}$ ($15,840\text{ pt}$)
    - **$31,680\text{ px}$ fits under the 32,000 px canvas limit**, while a full 15,000-character letter will only use 6 to 11 pages ($9,504\text{ px}$ to $17,424\text{ px}$).

I'd like to have the printable PNG for an archive that aligns the letters side by side. If a letter is omitted, an empty box where text could go is used. The subject isn't included. That could be useful for unwanted messages, especially from external sources.

The printable PNG for each letter can fit them into pages that are aligned side by side too.

There should be a printable option, so that a PDF can be obtained.

The full size of the document should be included in the PNGs, with the margins.

The side-by-side printable PNG engine and printable PDF options have been implemented according to your specifications.

Key Implementations

1. Standard Page Geometry at 144 DPI ($2\text{ px} = 1\text{ pt}$)

- **Full Document Size Included**:
    - Width: **612 pt (1,224 px)**
    - Height: **792 pt (1,584 px)**
    - Margins: **1 inch / 72 pt (144 px)** on all 4 sides included in the canvas.
- **Printable Text Area**:
    - Width: **468 pt (936 px)**
    - Height: **648 pt (1,296 px)**
    - **Border**: Rendered outside the text area, framing the $936\text{ px} \times 1,296\text{ px}$ text box inside the 1-inch margins.

- **Typography**:
    - Font: **12pt (24 px)** monospace (`"SF Mono", Menlo, Consolas, Courier New`).
    - Leading / Pitch: **18pt (36 px)** (1.5× leading).

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2. Archive Page PNG: Letters Aligned Side by Side

- Each letter on the archive page occupies a standard page slot aligned horizontally ($X = \text{slot} \times 1,224\text{ px}$).
- **Omitted Letters (Empty Box, No Subject)**:
    - If a letter is unchecked or omitted via the *"Omit External Letters"* option, an **empty framed box where text could go** is rendered.
    - **The subject is omitted entirely** (no title, no sender header, no body text), providing a clean blank slot.
- **Included Letters**:
    - Rendered with 12pt typography, header metadata, wrapped body lines, and footer archive status.
- Exported via `[ 📷 Export Archive PNG (Side-by-Side) ]`.

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3. Individual Letter PNG: Pages Aligned Side by Side

- When downloading an individual message via `[ 📷 Download As PNG ]`, long messages paginate across standard 8.5″ × 11″ pages placed **side by side** horizontally:
    - Page 1: $X = 0$ (includes letter header, metadata, and initial body lines).
    - Continuation Pages: $X = P \times 1,224\text{ px}$ (includes continuation header and subsequent body lines).
    - Canvas height: **1,584 px (792 pt)**; Canvas width: $\text{pages} \times 1,224\text{ px}$.
    - Every page preserves the full document size with 1-inch margins and outer border.

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4. Printable Option for PDF

- **Individual Letter PDF**:
    - Added a `[ 🖨️ Print / PDF ]` button to each message card footer across all tabs.
    - Clicking this opens a clean print layout formatted for **8.5″ × 11″ portrait with 1-inch margins, bordered text area, and 12pt font**, triggering the browser's native print dialog to immediately **"Save as PDF"**.

- **Archive Page PDF**:
  - The toolbar button `[ 🖨️ Print Archive / PDF ]` prints the archive page with omitted letters formatted as empty bordered boxes without subjects.

File Locations & Status

- **Codebase Updated**:
  - tricurse/contact/index.php
  - addressbar.net/contact/index.php
  - `/var/www/html/contact/index.php`
- **Validation**: PHP syntax (`php -l`) and JavaScript syntax (`node -c`) pass with 0 errors; changes committed to git on `main`.